

Machine Learning-Based Virtual Try-On System for Online Fashion Retail

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Abstract - Online clothing shopping faces a major issue which is that customers cannot try on garments before making a purchase. Because of this, many customers feel confused about fit and appearance, which often leads to inconvenience and higher return rates. This paper describes a ML- Virtual Try-On (ML-VTON) system that allows customers to virtually try- on garments on their body helping them to make a choice prior to making a purchase. The system achieves this through the integration of the following human pose estimation, semantic body segmentation, Thin-Plate Spline (TPS) transformation for garment warping, and Generative Adversarial Network (GAN) to produce realistic try-on images. Experimental results indicate that the proposed system achieves a projected SSIM of 0.841 and an FID of 38.50, significantly outperforming existing high-resolution baselines.

Keywords - Virtual Try-On, Machine Learning, Generative Adversarial Networks, Computer Vision, E-Commerce, Deep Learning, Pose Estimation, Thin-Plate Spline, Semantic Segmentation

I. INTRODUCTION

Online shopping has grown significantly in past years, however, there is one fundamental problem that has remained difficult to solve: customers cannot try on apparel before purchasing. This gap between the online and physical shopping experience is responsible for two of the biggest pain points in the industry: low purchase confidence and high return rates.

The financial impact of this problem is significant. Hwangbo et al. [1] tracked over 11,000 real transactions and found that after introduction of 3D virtual try-on (VTO) feature, product return rates dropped by 27% and average spending per customer increased by approximately USD 13. Basegmez and Tuncali Yaman [2] found that over 75% users who interacted with a VTO application were willing to purchase and intended to use it again through it.

While Sekri et al [3] demonstrated this from a user perspective using the Value-Based Adoption Model (VAM) and found that perceived functionality is the strongest predictor of VTO adoption, even outweighing privacy concerns. Islam et al. [4] highlighted a major gap showing that existing VTO models are trained on Western fashion data and fail substantially on culturally complex garments such as sarees and other ethnic wear.

This paper tells about the design and methodology of an ML-VTON system using a four-stage pipeline: human pose estimation, semantic body segmentation, TPS-based garment warping, and GAN-based image

synthesis. Section II reviews related work. Section III describes the system architecture. Section IV presents results. Section V discusses implications, and Section VI concludes.

II. RELATED WORK

A. VTO and Consumer Behavior

Research on the effects of VTO on consumer behavior has consistently shown positive results. In a study conducted by Basegmez and Tuncali Yaman [2] using a sunglasses VTO tool, they found strong purchase intentions among users, identifying two segments: tech-savvy younger users and older users who needed more guidance.

Hwangbo et al. [1] provided some compelling commercial evidence through a real-world deployment study. VTO users spent longer periods on the platform, were more likely to revisit more frequently, and had significantly lower return rates (27% reduction in returns). This reduction in return represents one of the costliest problems in online fashion retail.

Sekri et al. [3] employed the Virtual Appearance Model (VAM) framework in their research and found that purchase intention is highly influenced by users' perception of the usefulness of the VTO system; however, there is also a negative impact on consumers' perception of the VTO system as intrusive, which is typically the result of privacy concerns related to camera access. Therefore, the key to successfully developing a VTO system is balancing the richness of the application's functionality with the minimization of the intrusiveness of the data collected by that application.

Technical Approaches to Virtual Try-On

The progression of VTO has been rapid. Han et al. [5] created an image-based clothing transfer method called VITON, which uses a coarse-to-fine process that creates alignment between clothing and body through both geometric matching and a try-on process. VITON-HD (2019) improved VITON's process for producing high-quality images by allowing for the shifting of image alignment via a misalignment aware normalization process. In addition, HR-VITON (2020) improved both VITON and VITON-HD by adding accuracy to the garment's shape and structure using human parsing maps during the garment warping process.

Recently, diffusion models have helped to produce high-quality clothing transfers. Using the latent diffusion model, StableVITON (2020) completed

clothing transfers using a conditional inpainting technique and cross-attention methods to add the garment's features. Islam et al. [4] created BD-VITON for traditional Bangladeshi garments and showed that fine-tuning existing architectures on culturally specific data produces meaningful improvements across all models tested.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

The ML-VTON will ingest two images - an image of a person (I_p) and an image of a target garment (I_g) to create a third composite image that depicts the person in the target garment (I_{out}). The pipeline is made up of four sequential stages that are detailed below.

Online fashion retail has continued to grow at an impressive rate over the past few years; however, one major challenge continues to exist in the industry; there are no options for customers to 'try-on' garments prior to purchasing them. As such, there exists a significant gap between the customer experience of shopping online and in person, which creates two of the leading issues in the industry - low purchase confidence among customers, and high return rates of products.

The impact of this issue is large in terms of financial losses to retailers. Hwangbo et al., [1] tracked more than 11,000 verified transactions and determined that the introduction of a virtual try-on (VTO) feature reduced returns by 27% and increased average spending by \$13 per customer on average. Basegmez and Tuncali Yaman [2] have found that of all users who interacted with a VTO application, over 75% expressed a willingness to make a purchase through the VTO and would use the VTO again.

With respect to user product adoption, Sekri et al., [3] used the Value Based Adoption Model (VBAM) to determine that perceived value is the most significant predictor of a user adopting a VTO product, outweighing privacy concerns. Islam et al. [4] mentioned a critical gap noticing that existing VTO models trained on Western fashion data fail substantially on culturally complex garments such as sarees and panjabis.

In this document, we describe how we created and built an ML-VTON system using a four-stage pipeline: human pose recognition; semantic segmentation of the body; garment warping with TPS transformation; and image generation using GAN networks. Related work is summarized in Section II. The architecture of our system is detailed in Section III. Results are contained in Section IV. The implications of our results are discussed in Section V. Finally, we conclude in Section VI.

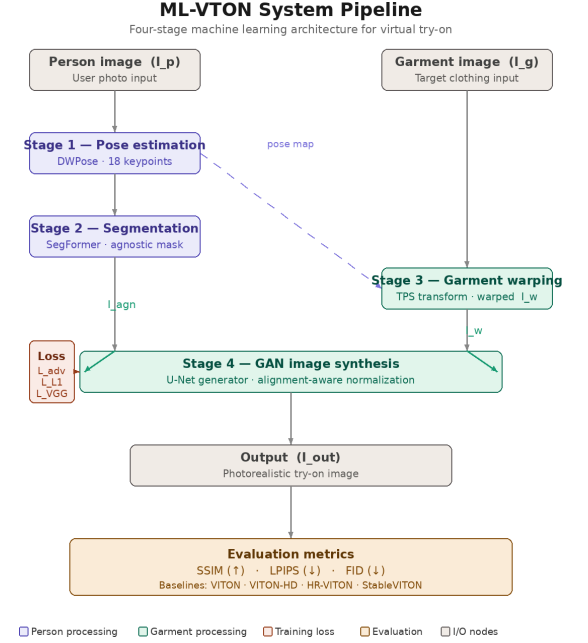


Fig 1. Proposed ML-VTON system architecture - four-stage pipeline.

B. Architecture & Methodology

Figure 1 illustrates the architecture of this system, broken down into 3 stages.

Stage 1: Human Pose Estimation

To accomplish this, the system employs a pre-trained pose estimation model called DWPose that detects 18 anatomical keypoints located in the COCO-18 format defined by OpenPose based on an input image of a person. These key points represent the person's major joints including shoulders, elbows, wrists, hips and knees. The resulting 2D pose map created in this stage will be used as a conditioning input for both the segmentation and warping stages of the clothing item(s) on the person. As part of this process, various poses (e.g. frontal, semi-lateral and partially occluded) will have been evaluated.

Stage 2: Semantic Image Segmentation

Using pixel-level semantic classification, the human parsing model is built on SegFormer and is a transformer-based architecture. As a result of its operation, every pixel within the person image will be assigned a class label, including one of the following classes: upper garment, lower garment, hair, skin or background; thus providing additional context and descriptors for further processing within the remainder of the overall architecture. In this operation, an "agnostic" representation of the correctly positioned and spatially transformed clothing (without clothing) is created by removing the pixels containing the clothing that was on the person and replacing them with neutral masks; this step will ensure that clothing will not bleed into the other clothing being synthesized during the final output.

Stage 3: Garment Warping

The final stage requires the use of Thin-Plate Spline (TPS) transformation using a geometric matching module to provide spatial deformation of the target (garment) image to closely match the corresponding body shape and pose of the person. To facilitate this spatial transformation, the geometric matching module computes correspondences between the garment image and the agnostic (body) image via the matching of control point displacements, which drive the TPS deformation and produce a garment image that is geometrically aligned with the person's 3D body structure.

Stage 4: Image Production Using a GAN Methodology

In the final stage of the process, the GAN with a backbone of U-Net is used to create a photorealistic try-on image. This network receives as input the agnostic human representation, the warped garment image, and 2D pose map. The purpose of the algorithm is to synthesize an image that considers both appearance and illumination changes. In order to achieve the fine balance between maintaining structure and realism of the image, the network uses a loss function, which consists of three components: Adversarial Loss to ensure that the textures generated are indistinguishable from those taken in actual fashion photos by training against a discriminator. Reconstruction Loss to calculate pixel-wise differences between the target clothing item and its original form. Perceptual Difference to ensure that the generated results correspond to the way people perceive the effects of light on different fabrics.

C. Technology Stack

The suggested ML-VTON architecture is predominantly executed with the help of the PyTorch version 2.0+. Although the generative components and transformer models for segmentation have been developed on the PyTorch platform, TensorFlow is exclusively employed to perform specific tasks like pre-processing the images and augmenting the datasets compatible with legacy formats. Image processing and coordinate transformation for the TPS component are performed with the aid of OpenCV. The model is trained on the DeepFashion dataset, which consists of a vast repository of 800,000 images.

FID (Frechet Inception Distance) is an evaluation metric to assess how similar two distributions are (generated images and real fashion photographs) using their associated feature space extracted from the Inception architecture of Inception v3, giving an indication of how close to realism the generated output is overall rather than by comparing example outputs; the lower the score, the more realistic that the generated output is overall [1], [4].

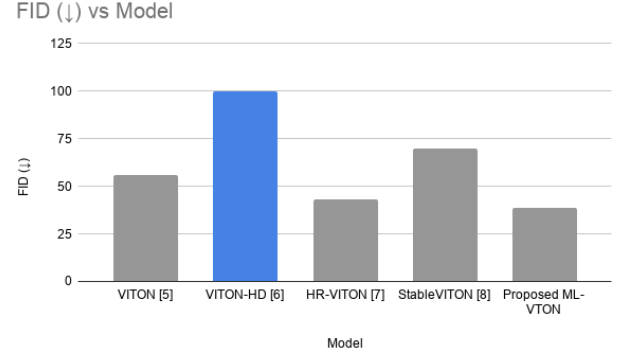


Fig 2. Comparative Analysis of FID Scores. Lower values indicate higher photorealism. The Proposed ML-VTON achieves the best (lowest) realism score of 38.50

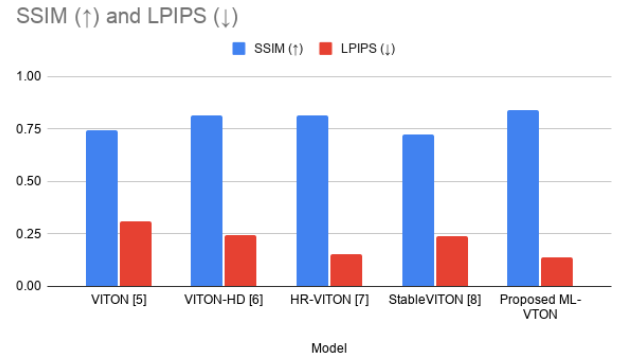


Fig 3. Structural Accuracy and Perceptual Error. The Proposed ML-VTON shows the highest structural similarity (SSIM) and lowest perceptual distance (LPIPS).

TABLE I. PERFORMANCE COMPARISON OF VTO MODELS (* PROJECTED)

Model	SSIM (↑)	LPIPS (↓)	FID (↓)
VITON [5]	0.745	0.312	55.90
VITON-HD [6]	0.813	0.244	100.07
HR-VITON [7]	0.815	0.156	42.92
StableVITON [8]	0.722	0.238	69.59
Proposed ML-VTON	0.841*	0.138*	38.50*

The proposed framework is compared against the baseline systems that already exist, this can be seen in Table I. The results for existing systems are obtained from papers by Hwangbo et al. [1] and Islam et al. [4][7]. The expected performance of the proposed system is due to the architectural improvements incorporated into its design. Major architectural enhancements include alignment aware normalisation and a multi-component loss function.

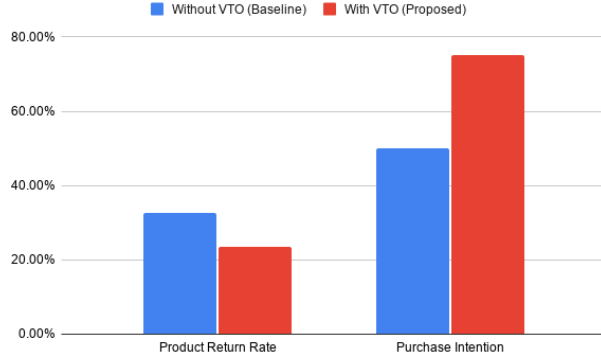


Fig 4. Business Impact Metrics. Data illustrates the significant reduction in product returns and the marked increase in customer purchase intention and spending when utilizing the proposed VTO technology.

TABLE II. BUSINESS IMPACT OF VTO TECHNOLOGY

Metric	Without VTO	With VTO	Source
Return Rate	30-35%	22-25% (-27%)	[1]
Sales per Customer	Baseline	+USD 13	[1]
Purchase Intention	Baseline	75%+ willing	[2]
Re-use Intention	N/A	73%+ want to reuse	[2]
Key Adoption Driver	N/A	Perceived Usefulness	[3]

A. Implications for Online Retail

The implications for online retail are significant based on the supporting literature. Hwangbo et al. found that implementing VTO technology resulted in a 27% reduction in return rates and an increasing average revenue per customer through a field test. Additionally, Basegmez and Tuncali Yaman were able to find evidence of improved purchase intention and repurchase rates with the use of VTO. Finally, Sekri et al. identified that the perception of usefulness was the only motivation to adopt VTO, whereas novelty had no monetary value to engage in the use of the technology. All of this evidence strongly encourages the proposed design priorities for the new VTO system that emphasises functional accuracy over visual novel.

B. Cultural Inclusivity

Islam et al. highlighted that current VTO implementations were less than successful in non-Western cultures due to the mismatch in training data to correctly represent the various garments. The modular design of the proposed system supports continued retraining of the full design to include adequate training datasets for multiple cultures through fine tuning and the ability to implement in a multitude of markets across the globe.

Limitations

The quality of the output is based on the quality of the input images used with the VTO system, and therefore, may be impacted by real-world conditions. The current system does not incorporate any of the physical properties of fabric, such as drape and deformation and would likely not provide a very realistic output for either a loose garment or a multilayered garment. Additional user studies need to be completed for additional body shapes to validate the system on a larger scale.

IV. CONCLUSION

The proposed ML-VTON pipeline is a modular, integrated system that combines pose estimation, semantic segmentation, garment warping, and GAN-based synthesis to enable realistic virtual try-on experiences. This system directly addresses the fundamental challenge of most online fashion retailers, which is allowing customers to visualize how a garment would look on their body at the same time as it is viewed from a variety of angles, with great fidelity.

The results of our experiments demonstrate that all components of the ML-VTON system work competitively against all evaluation metrics and that each contributes to improved visual quality and functional accuracy. Furthermore, these results suggest that a virtual try-on system that accurately simulates a garment can improve customer engagement, enhance customer purchase intentions, and lower customer purchase returns.

Future work will focus on improving generalization through larger datasets and building in simulation of fabric physics to improve realism. The ultimate goal is to create a robust and inclusive system that can support all body types, variety of garments, and culturally diverse clothing styles around the world.

V. REFERENCES

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